Advanced Flows

Lecture 9 | Continuous Flows and Flow Matching

Ojas Shirekar | 26th May 2026

Remember?

Normalising Flows

$$x = g(z, w), \qquad z = g^{-1}(x, w)$$

A normalising flow learns an invertible transformation from simple noise to data.

$$g = g_K \circ \cdots \circ g_1(z)$$

$$p_{\mathbf{X}}(\mathbf{x}) = p_{\mathbf{Z}}\left(g^{-1}(\mathbf{x})\right) \left| \det J_{g^{-1}}(\mathbf{x}) \right|$$

What *changes* today?

Out with the old, in with the new (?)

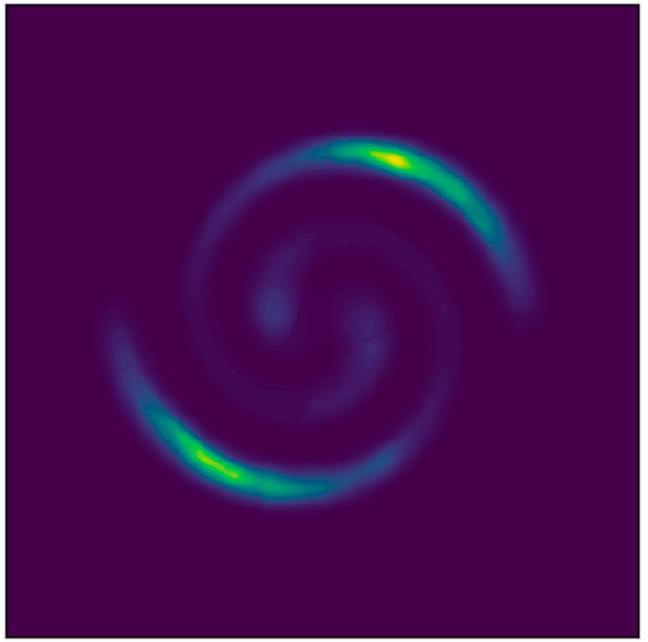
$$g = g_K \circ \cdots \circ g_1(z)$$

Instead of composing many discrete invertible layers, we define a continuous flow through a *velocity field*.

$$\frac{dx_t}{dt} = v_\theta\left(x_t, t\right)$$

Generative Models

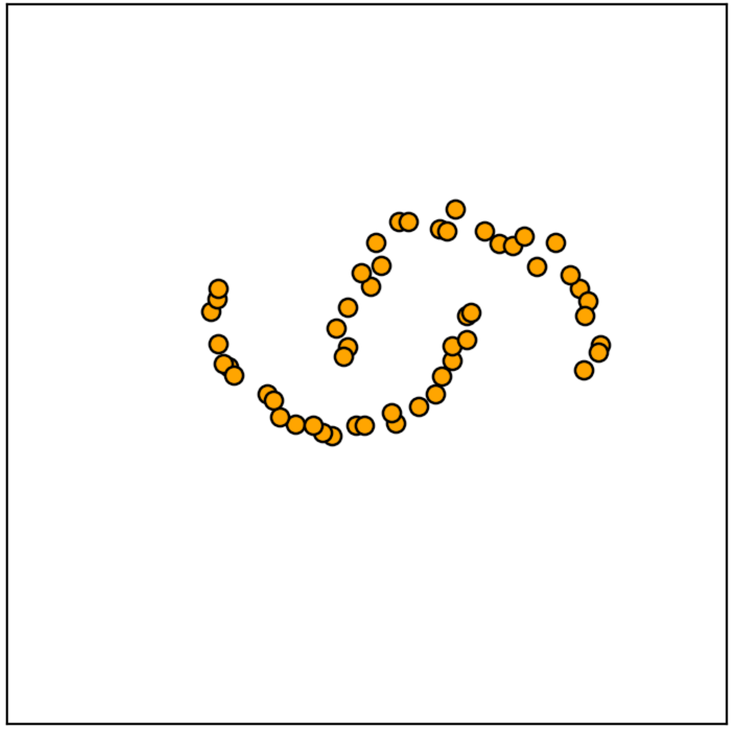
- Unknown: data distribution q



Generative Models

- Unknown: data distribution q
- Given: samples $x_1 \sim q$

$x_1 \sim q$



Sampling

$x_0 \sim p$

$x_1 = \psi_\theta(x_0)$

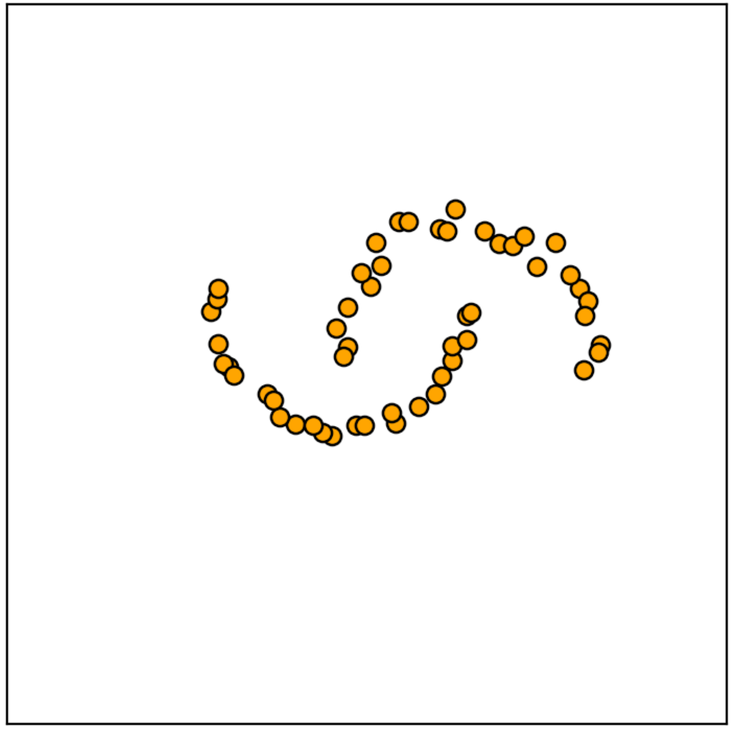
Density Estimation

$p_\theta \approx q$

Generative Models

- Unknown: data distribution q
- Given: samples $x_1 \sim q$

$x_1 \sim q$



Generative Model

(ψ_θ, p_θ)

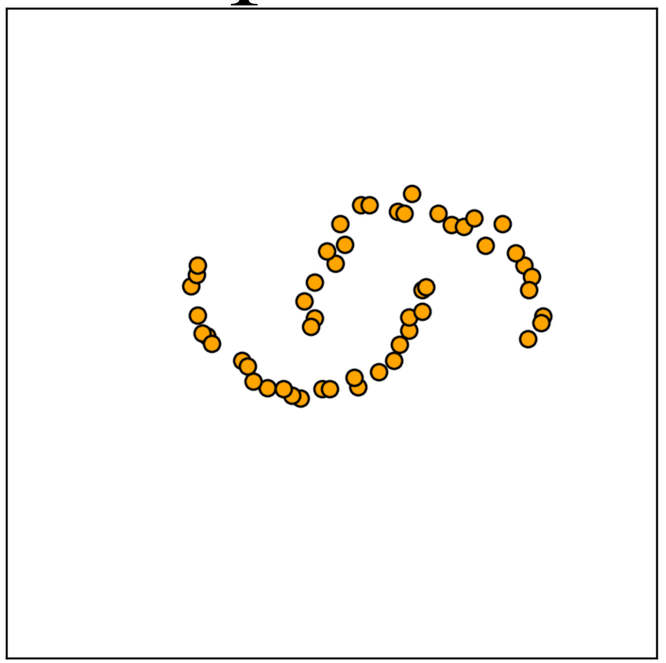
Generator

Underlying Density

Generative Models

- Unknown: data distribution q
- Given: samples $x_1 \sim q$

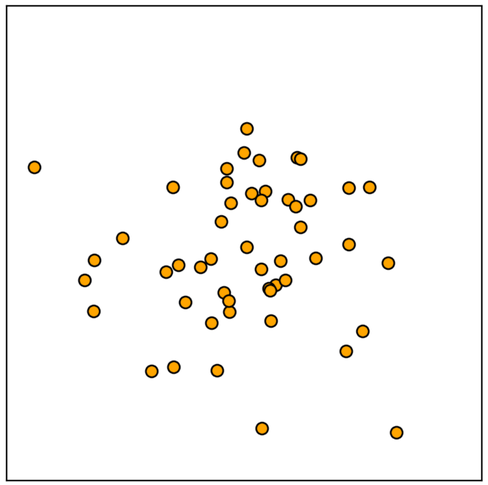
$x_1 \sim q$



Goal: find parameters θ s.t. $p_\theta \approx q$

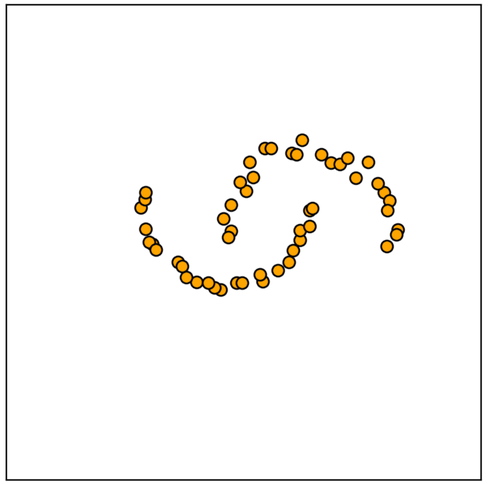
Generative Models

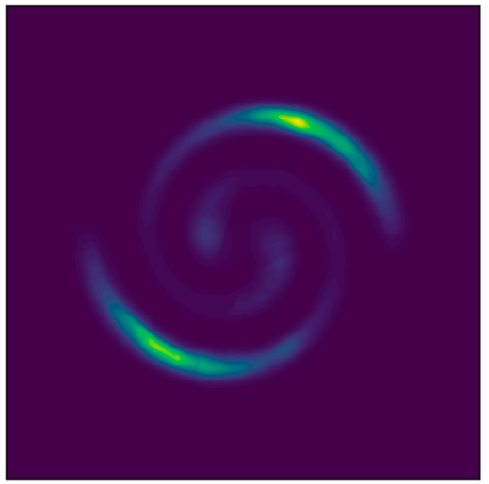
$x_0 \sim p$



ψ_θ

$x_1 \sim q$





Sampling

$x_0 \sim p$

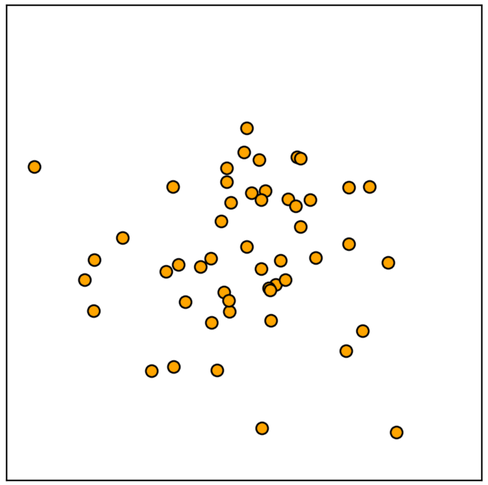
$x_1 = \psi_\theta(x_0)$

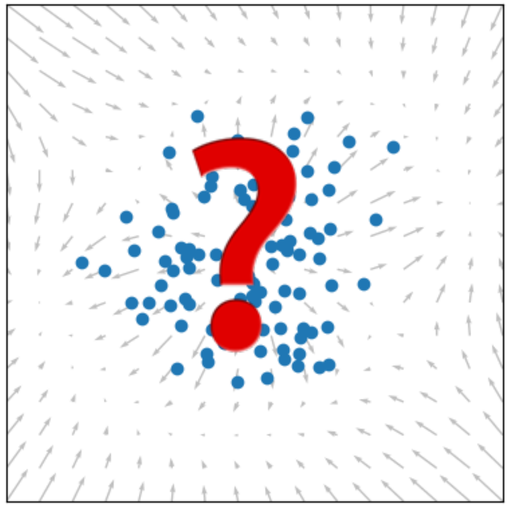
Density Estimation

$p_\theta \approx q$

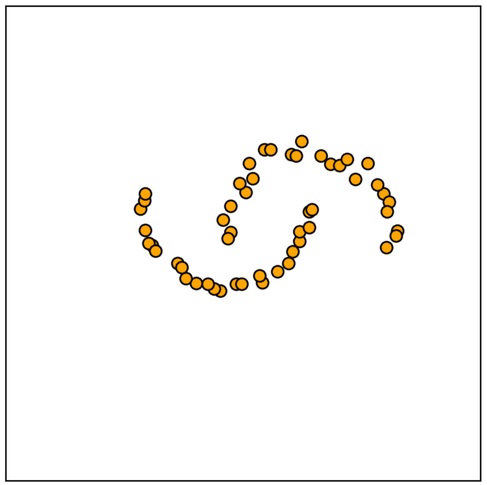
Continuous Normalising Flows

$x_0 \sim p$





$x_1 \sim q$



$$\psi_t : [0,1] \times \mathbb{R}^2 \to \mathbb{R}^2$$

$$\frac{dx_t}{dt} = v_t(x_t)$$

Neural Ordinary Differential Equations (2018)

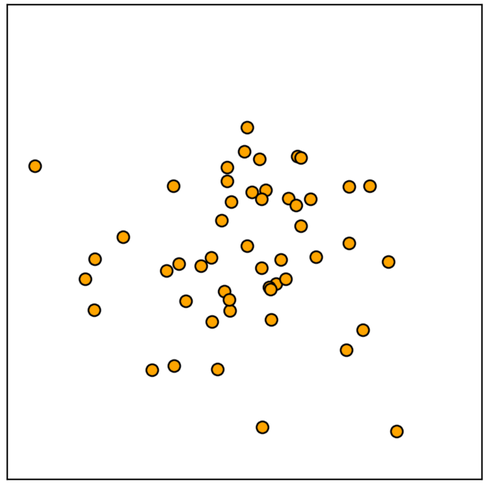
Continuous Normalising Flows

A Dynamical View

Flow ODE

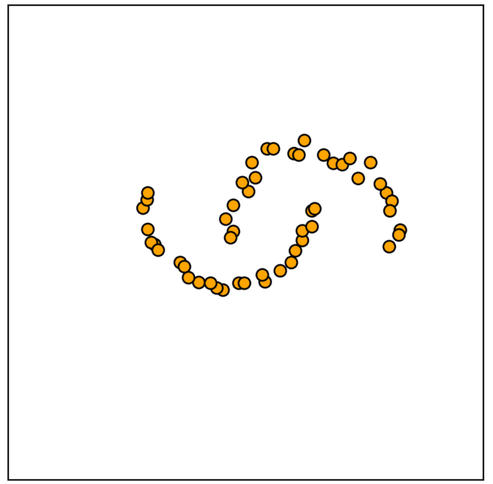
$$\dot{\psi}_t(x_0) = v_t\left(\psi_t(x_0)\right)$$

$x_0 \sim p$



ψ_1

$x_1 \sim q$



$$\psi_t : [0,1] \times \mathbb{R}^2 \to \mathbb{R}^2$$

Velocity Field $\quad v_t : [0,1] \times \mathbb{R}^2 \to \mathbb{R}^2$

Continuous Normalising Flows

A Dynamical View

$$\dot{\psi}_t(x_0) = \frac{d}{dt}\psi_t(x_0)$$

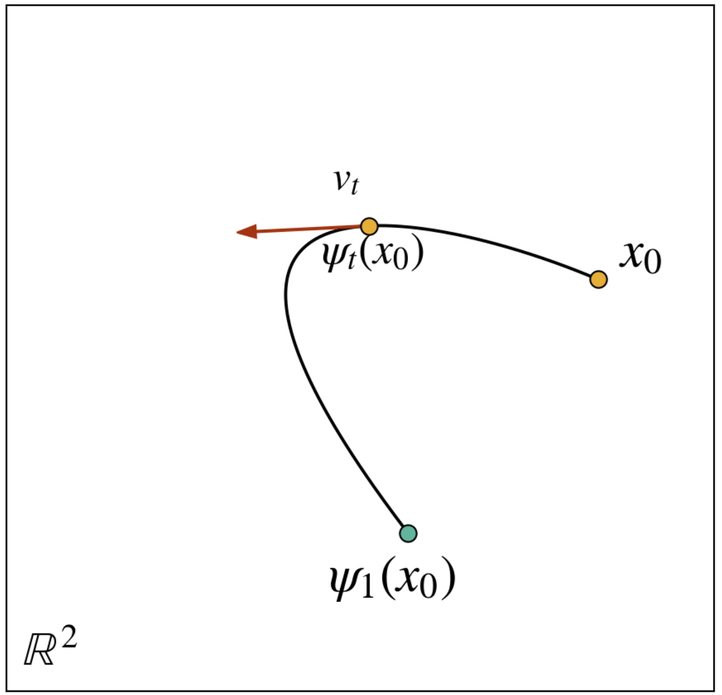
Flow ODE

$$\dot{\psi}_t\left(x_0\right) = v_t\left(\psi_t\left(x_0\right)\right)$$

So the full story is

Location: $x_t = \psi_t(x_0)$

Velocity: $\dot{x}_t = v_t(x_t)$



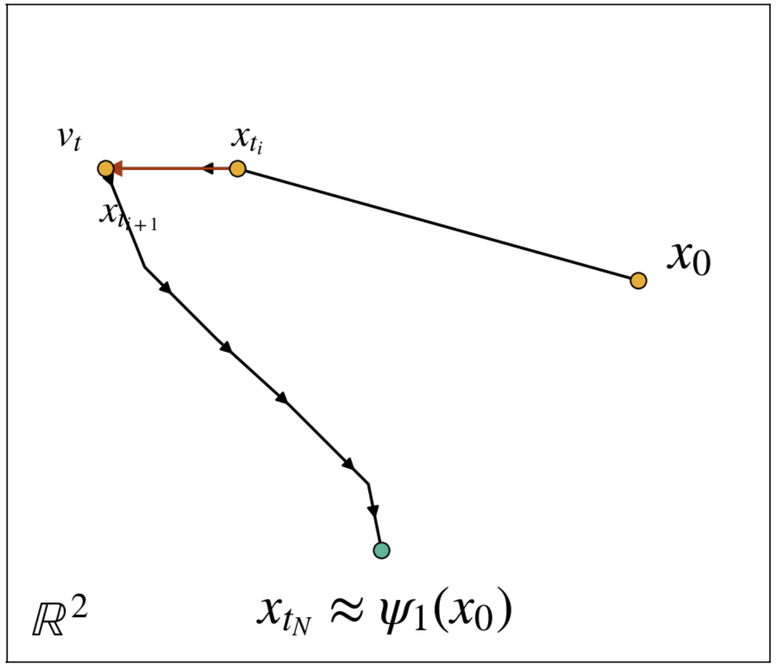
$$\psi_t\left(x_0\right) = x_0 + \int_0^t v_s\left(\psi_s\left(x_0\right)\right) ds$$

Continuous Normalising Flows

A Dynamical View

Flow ODE

$$\dot{\psi}_t\left(x_0\right) = v_t\left(\psi_t\left(x_0\right)\right)$$



Sampling

e.g., Euler Solver:

$$x_{t_{i+1}} = x_{t_i} + \Delta t \cdot v_t\left(x_{t_i}\right)$$

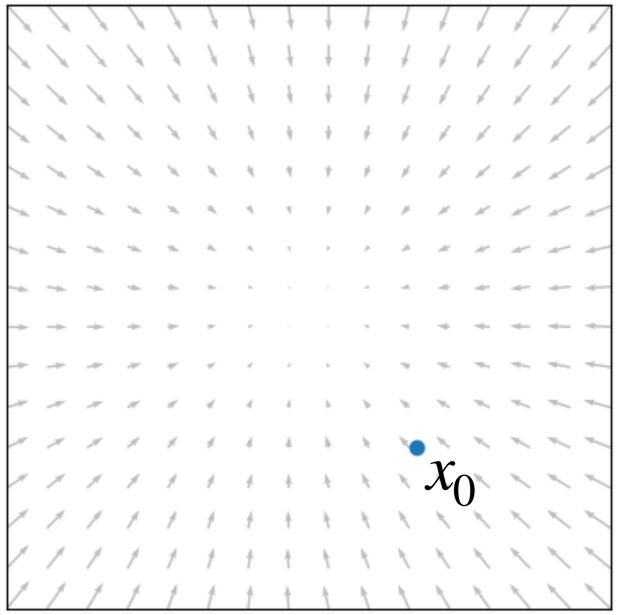
$$N = \frac{1}{\Delta t}$$

Continuous Normalising Flows

A Dynamical View

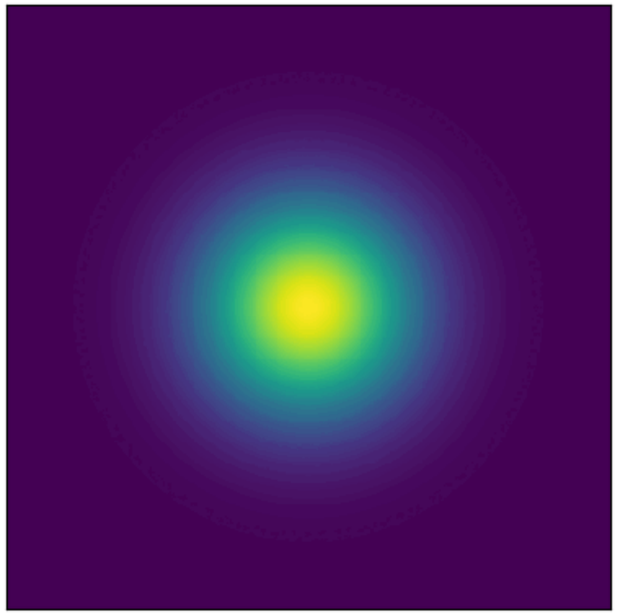
Flow ODE

$$\dot{\psi}_t(x_0) = v_t\left(\psi_t(x_0)\right)$$



Continuity Equation PDE

$$\dot{p}_t = -\operatorname{div}(p_t v_t)$$



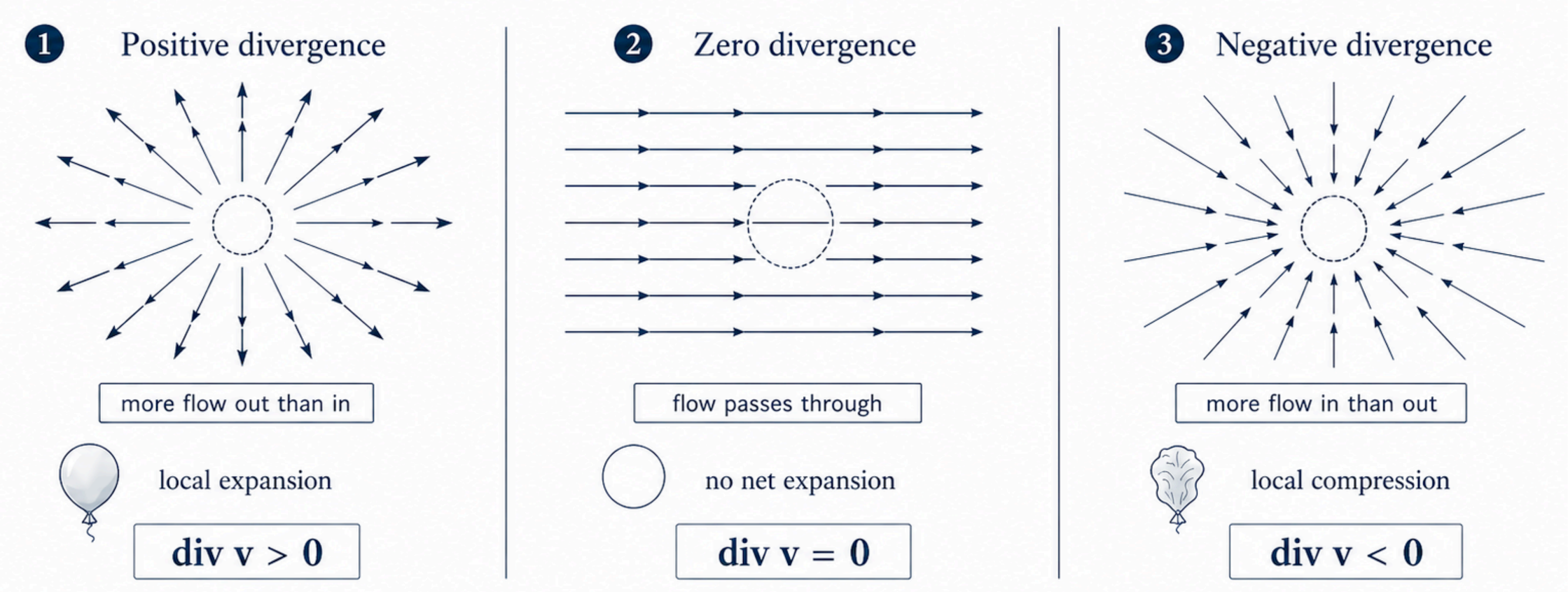
v_t generates p_t iff the continuity equation holds $\forall t \in [0,1], \quad x \in \mathcal{X}$

What's div?

With velocity field $v(x) = \begin{bmatrix} v_1(x_1, x_2) \\ v_2(x_1, x_2) \end{bmatrix}$

The **divergence** is given by $\nabla \cdot v = \frac{\partial v_1}{\partial x_1} + \frac{\partial v_2}{\partial x_2}$

What's div?


1
Positive divergence
more flow out than in
local expansion
div v > 0
2
Zero divergence
flow passes through
no net expansion
div v = 0
3
Negative divergence
more flow in than out
local compression
div v < 0

Continuity and Flow ODE

Constant velocity field example

$$v_t(x) = v(x, t)$$

$$\frac{dx_t}{dt} = v_t(x_t)$$

$$v_t(x) = a$$

$$x_t = x_0 + t \cdot a$$

$$\psi_t(x_0) = x_0 + t \cdot a$$

$$\nabla \cdot v_t = 0$$

Remember? 💡

Normalising Flows

$$x = g(z)$$

$$J_g(z) = \frac{\delta g}{\delta z}$$

new vol $\approx |\det J_g(z)| \times$ old vol

$$p_X(x)\,dx = p_Z(z)\,dz$$

Continuous Normalising Flows

$$\frac{dx_t}{dt} = v_\theta\left(x_t, t\right)$$

we are not applying one big map all at once

$$\nabla \cdot v_t(x)$$

At this **instant**,

is the **local cloud expanding or contracting**?

$$\partial_t p_t(x) = -\nabla \cdot \left(p_t(x)\, v_t(x)\right)$$

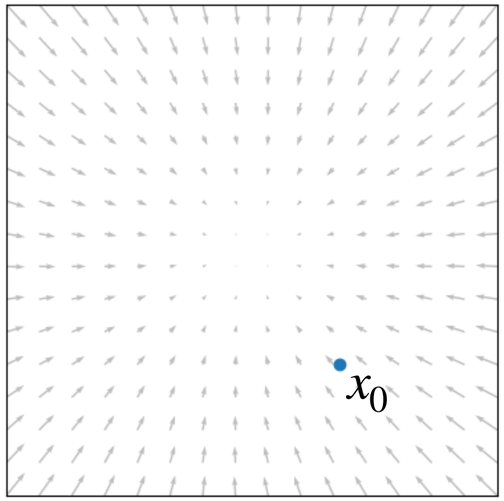
The density at a point changes depending on how probability mass flows into or out of that point.

Continuous Normalising Flows

A Dynamical View

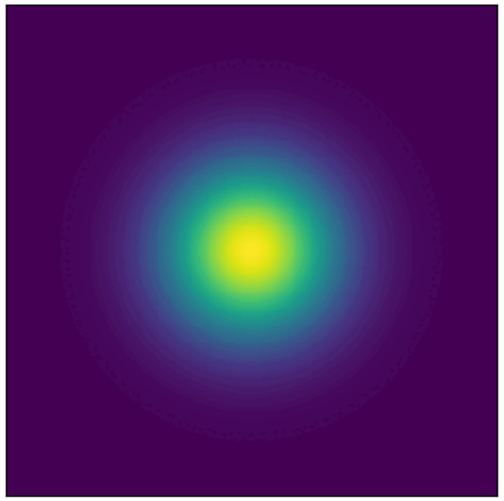
Flow ODE

$$\dot{\psi}_t(x_0) = v_t\left(\psi_t(x_0)\right)$$

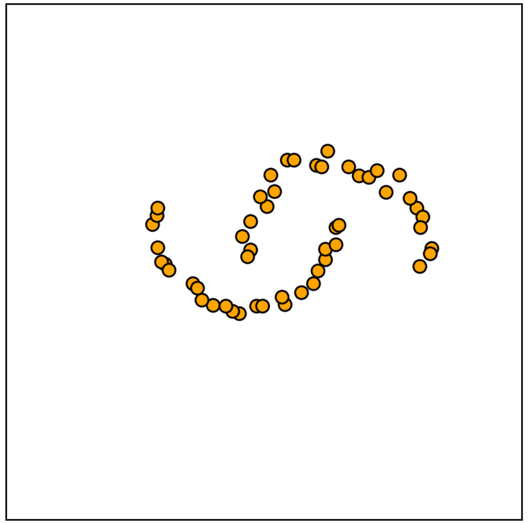


Continuity Equation PDE

$$\dot{p}_t = -\operatorname{div}(p_t v_t)$$



Given: samples $x_1 \sim q$

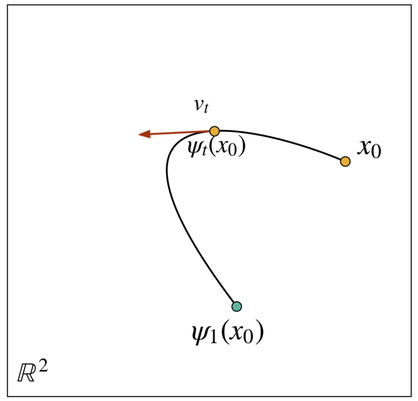


Goal: find velocity field $v_t(x)$ s.t. $p_1 \approx q$

How do I train one?

Continuity Equation PDE

$$\dot{p}_t = -\operatorname{div}(p_t v_t)$$



Instantaneous change of coordinates (trajectory x_t)

$$\frac{d}{dt}\log p_t\left(x_t\right) = -\operatorname{div}\left(v_t\left(x_t\right)\right)$$

Log-likelihood computation

$$\log p_1\left(x_1\right) = \log p\left(x_0\right) + \int_1^0 \operatorname{div}\left(v_t\left(x_t\right)\right) dt$$

$$x_t = x_1 + \int_1^t v_s\left(x_s\right) ds$$

How do I train one?

- Simulating x_t
- Backprop through simulation
- Estimate $\mathrm{div}(v_t)$
- Compute $\log p(x)$

Log-likelihood computation

$$\log p_1\left(x_1\right) = \log p\left(x_0\right) + \int_1^0 \mathrm{div}\left(v_t\left(x_t\right)\right) dt$$

$$x_t = x_1 + \int_1^t v_s\left(x_s\right) ds$$



$$D_{\mathrm{KL}}\left(q \| p_1\right) = -\,\mathbb{E}_{x\sim q} \log p_1(x) + c$$

Now, Flow Matching

Learn the velocity of a chosen path

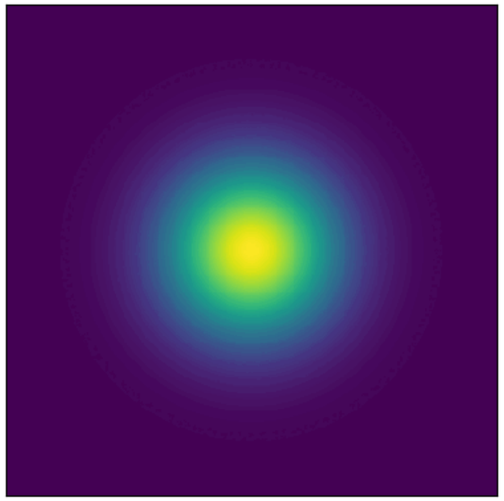
$$D_{\mathrm{KL}}\left(q \| p_1\right) = -\mathbb{E}_{x\sim q} \log p_1(x) + c$$

Infinitely many probability paths p_t

Infinitely many velocity fields v_t

Continuity Equation PDE

$$\dot{p}_t = -\operatorname{div}(p_t v_t)$$





Now, Flow Matching

Learn the velocity of a chosen path

Let us choose a path of distributions p_t, p_0 is noise

$$p_0 = p, \qquad p_1 \approx q$$

$$\frac{dx_t}{dt} = u_t(x_t)$$

Assume there is a correct velocity per-sample

$$v_\theta(x, t) \approx u_t(x)$$

$$\mathscr{L}_{FM} = \mathbb{E}_{t,x\sim p_t}\left[\left\| v_\theta(x, t) - u_t(x) \right\|^2 \right]$$

Now, Flow Matching

Learn the velocity of a chosen path

Let us choose a path of distributions p_t, p_0 is noise

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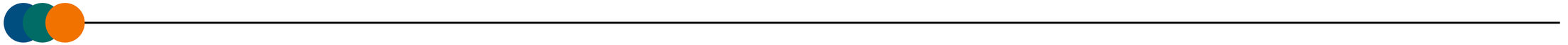
At random time t, look at a random point x on the path and ask the net:

Which way should this point move?

Now, Flow Matching

Learn the velocity of a chosen path

$$x_0 \sim p, \qquad x_1 \sim q$$



x_0 x_t x_1

noise time t data

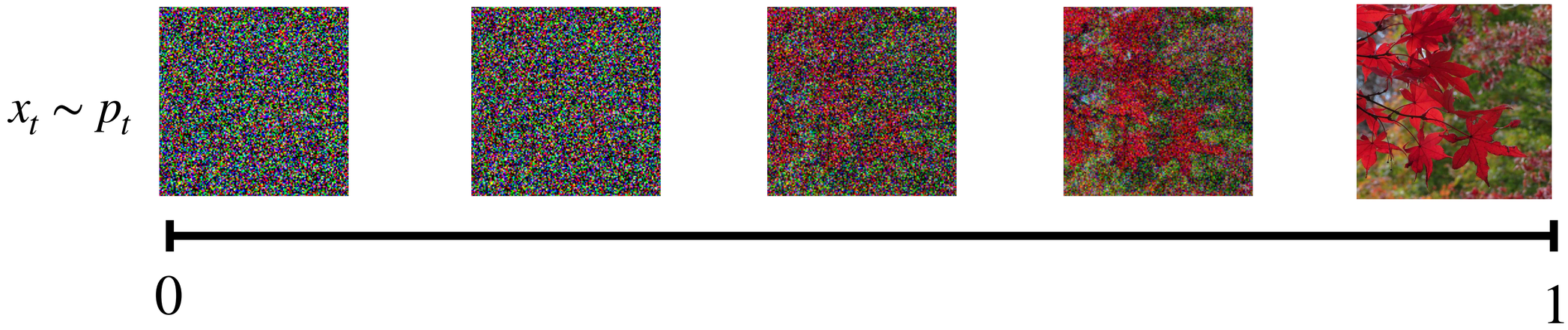
Straight-line interpolation

$$x_t = (1-t)x_0 + tx_1$$

$$u_t = \frac{dx_t}{dt} = x_1 - x_0$$

Training is supervised regression on velocities! 🥺 $\mathscr{L}_{FM} = \mathbb{E}_{t,x\sim p_t}\left[\left\| v_\theta(x,t) - u_t(x) \right\|^2 \right]$

Key Insight



The neural network does not memorize pairwise lines. It learns the average velocity needed at each location and time.

Key Insight

A CNF asks:

“Move particles from noise to data, and keep track of how the density expands or compresses.”

Flow matching asks:

“Here are examples of how particles should move from noise to data; learn that velocity field.”

NF, CNF, Flow Matching and Diffusion

Model	Object learned	How probability moves	Training	Sampling
Normalising Flow	Invertible map g_θ	Finite transformation $z \mapsto x$	MLE using change of variables and Jacobian determinant	Direct: sam p_Z, then com $g_\theta(z)$
Continuous Normalising Flow	Velocity field $v_\theta(x,t)$	Deterministic ODE flow	MLE using continuous change of variables and divergence	Solve $\frac{dx_t}{dt} = v$
Flow Matching	Velocity field $v_\theta(x,t)$	Deterministic ODE flow along a chosen path	Regression onto target velocities u_t	Solve ODE fr to data
Diffusion / Score Model	Score, noise, or denoising network	Stochastic noising, then learned reverse process	Denoising or score matching across noise levels	Reverse Sl probability-fl

NF, CNF, Flow Matching and Diffusion

One-line intuition for each model family.

Model	Slogan
Normalising Flow	Learn an invertible map.
Continuous Normalising Flow	Learn a continuous-time flow by likelihood.
Flow Matching	Learn the velocity of a chosen transport path.
Diffusion / Score Model	Learn how to reverse noise corruption.

Recap

- Flow ODE
- Continuity Equation
- CNF objective
- NF vs CNF
- Flow Matching objective

Further Reading

- https://diffusionflow.github.io/
- https://diffusion.csail.mit.edu/2026/docs/lecture_notes.pdf
- https://arxiv.org/abs/2412.06264
- https://arxiv.org/abs/1806.07366